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Sedimentology, Stratigraphy and Paleoenvironments of the Albian/Cenomanian Carbonates of the Arabian Platform, Southern Margin of the Tethys. A Platform-Scale Perspective

M. Deville de Periere, O. Alajaji, H. Boukhamsin, W. Basrawi, I. Hilali, F. Lu, A. Henderson, I. Mikellidou, F. De Ville de Goyet, and M. Soua, Saudi Aramco, Dhahran, Saudi Arabia

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Abstract

The Wasia Formation was deposited on the southern margin of the Shilaif Basin and comprises four 3rd order sequences and seven members (Khafji, Safaniya, Mauddud, Wara, Ahmadi, Rumaila, and Mishrif, in stratigraphic order). This study aims to refine the paleoenvironments of the Wasia Formation in the studied area, and to provide a detailed stratigraphic framework of the Albian–Cenomanian strata linked to the regional understanding of the Wasia Formation. Detailed sedimentological description and high-resolution bulk carbon isotope ($\delta^{13}\text{C}$) analysis, along with biostratigraphy, were performed in cores and sidewall cores from 17 wells from the Rub al Khali Basin. The biostratigraphic analysis includes micropaleontology (using both thin sections and disaggregated material), palynology (dynocysts and pollens) and nannofossils characterization. It is the first time that such an integrated dataset is presented for a single location at the scale of the Arabian Platform, which enables improved platform-scale correlations.

The first sequence marks the initial Albian flooding of the platform with the argillaceous, shallow marine to estuarine deposits of the Khafji Member. The overlying Safaniya Member contains orbitolinid-rich limestones and deep marine deposits, with planktonic foraminifera. This marks a regional flooding event associated with slightly anoxic conditions (intraself basin). The top of the Safaniya Member marks the Albian/Cenomanian transition according to planktonic foraminifera, which differs from previously established stratigraphy. This sequence changes upwards to shallower deposits of the early Cenomanian Mauddud Member, which is capped by rudist-rich deposits that have been affected by a regional unconformity. The middle Cenomanian section includes the clastic-rich Wara Member and the carbonates of the Ahmadi and Rumaila members, characterized by shallow marine low-energy deposits rich in diversified benthic foraminifera and green algae, with sporadic rudist developments. Boundaries between these members are associated with terrigenous layers at sequence boundaries. The last 3rd order sequence, consisting of the mid to upper Cenomanian Mishrif Member, contains abundant planktonic foraminifera and coccoliths indicating the drowning of the platform and the development of the Shilaif Basin associated with the OAE2. Biostratigraphy and carbon isotopes indicate that the uppermost part of the Turonian deposits are missing, eroded by the pre-Aruma unconformity.

This integrated study unravels paleoenvironmental changes on the southern margin of the Shilaif Basin and provides a new detailed stratigraphic framework for the deposition of these marine carbonates, in line with the regional understanding of the Wasia Formation on the Arabian Platform.

Introduction

The Cretaceous equatorial carbonate platforms, such as those on the Neotethys margins, are mainly characterized by shallow marine carbonate with interbedded intrashelf basin facies. These environments and associated biota are sensitive to sea-level fluctuations, climate changes, and carbon cycle perturbations, such as global oceanic anoxic events (OAEs). The Albian to Turonian period is hence critical due to drastic paleoenvironmental changes, including the OAE1d (Late Albian) and OAE2 (Cenomanian/Turonian). On the eastern Arabian Platform, Albian to Turonian strata, including the Natih Formation in Oman, Shilaif Formation in the UAE and Wasia Formation in Saudi Arabia, provide insights into the region's stratigraphy and paleogeography (e.g., [Alsharhan & Nairn, 1988](#); [Burchette, 1993](#); [Philip et al., 1995](#); [Scott et al., 2000](#); [van Buchem et al., 2002](#); [Grelaud et al., 2006](#); [Al-Zaabi et al., 2010](#); [Hollis, 2010](#); [Razin et al., 2010, 2017](#); [Taher et al., 2010](#); [Al Balushi & Macquaker, 2011](#); [Vahrenkamp, 2013](#); [Vahrenkamp et al., 2015](#); [Wohlwend et al., 2016](#); [Perrotta et al., 2017](#); [Hennhofer et al., 2019](#)). Despite extensive literature, biostratigraphy is mainly based on ammonite data with sporadic publications on detailed foraminifera assemblages ([van Buchem et al., 2011](#); [Bromhead et al., 2022](#)). In addition, there is a lack of integrated biostratigraphy data on multiple fossil groups including planktonic foraminifera, palynomorphs and nannofossils. This study presents a first comprehensive analysis of the Albian/Turonian Wasia Formation in the Rub' Al Khali Basin, Saudi Arabia, integrating detailed biostratigraphy results from several fossil groups, including benthic and planktonic foraminifera, palynomorphs, and nannofossils, calibrated with carbon isotopic data. The aim of the study is to provide a clearer understanding of stratigraphic and biota modifications over time on the eastern Arabian Platform.

Geological background

During the Albian/Turonian period, the Arabian Platform was located near the equator and dominated by a wide, shallow marine carbonate-dominated platform. During the Cenomanian, a wide intrashelf basin called the Shilaif Basin developed on the eastern part of the platform due to differential aggradation rates, locally associated with anoxic conditions and organic matter preservation ([Vahrenkamp et al., 2015](#); [Wohlwend et al., 2016](#); [Hennhofer et al., 2019](#); [Soua et al., 2024](#) – [Figure 1](#)). This basin had deep marine deposits with organic-rich sediments, abundant nannofossils, and locally common ammonites, while the basin margins were associated with large rudist biostromes and higher energy conditions developed in front of shallow marine deposits including wackestones and packstones hosting diverse benthic foraminifera, green algae, bivalves, and gastropods. In the Rub al Khali Basin (Saudi Arabia), the Wasia Formation includes six distinct lithostratigraphic units, which can be correlated with Oman's Natih Formation. These units include the Safaniya, Mauddud and Wara members (equivalent to Natih E, F and G), the Ahmadi and Rumaila members (equivalent to Natih D and C) and the Mishrif Member (equivalent to Natih B/A). The base of the Wasia Formation also encompasses the clastic-dominated Khafji Member, although this unit is beyond the scope of this study. The top of the Wasia Formation is affected by a major regional unconformity, related to a regional sea-level fall and local tectonic uplifts, known as the Pre-Aruma Unconformity, dated from the middle Turonian (K150 – [Brohmead et al., 2022](#); [Simmons et al., 2024](#)). This unconformity is locally associated with karstic structures, paleosoils, and meteoric diagenesis (e.g. [Burchette & Wright, 1992](#); [Hollis, 2010](#); [Deville de Periere et al., 2017](#)). The studied area is also affected by a Base Tertiary Unconformity, which eroded the Wasia Formation and is overlain by the Paleocene Shammar Member of the Umm er Radhuma Formation.

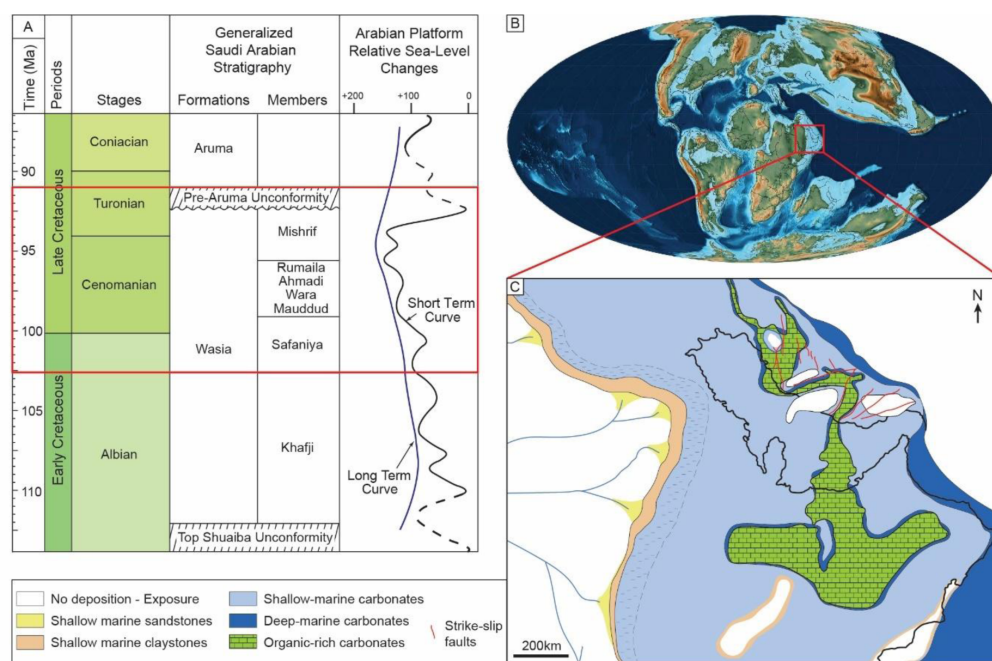


Figure 1—(A) Chronostratigraphy of the Albian to Coniacian including eustatic variations inferred for the Arabian Platform (modified from [Haq & Al-Qahtani, 2005](#)). The red box indicates the stratigraphic distribution of the Wasia Formation and internal subdivisions. (B) Global paleogeographical of the Cenomanian ([Scotese, 2014](#)). (C) Paleoenvironments of the Arabian Platform during the maximum flooding of the Late Cenomanian (modified from [Bromhead et al., 2022](#)).

Material and methods

The present study is based on the integration of 20 wells, including the detailed sedimentological description of approximately 320m of slabbled cores, along with 701 side wall cores (SWCs). The vertical understanding of the stratigraphy is based on synthetic sections built from lateral correlations using gamma-ray logs (GR) and available core data. Multi-scale observations of vertical and lateral facies changes, allochem diversity, sedimentary structures and key surfaces have been used to define depositional environments and to build the sequence stratigraphic framework. In addition to core observations, 632 thin sections have been prepared to refine facies identification, calibrate core descriptions and provide a detailed characterization of the flora-faunal assemblages for paleoenvironment and biostratigraphy purposes. Drill cuttings were locally used to fill sample gaps, especially in intervals not covered by core or SWCs. Argillaceous and/or organic-rich beds in core were systematically sampled for palynological (293 samples) and nannopaleontological (417 samples) analyses with the aim of providing a refined age dating of the Wasia Formation.

The mineralogical assemblage was studied through 407 rock chips selected for bulk X-ray diffraction (XRD) analysis. Ultimately, bulk stable isotope analysis ($\delta^{18}\text{O}$ and $\delta^{13}\text{C}$) were conducted on 250 samples selected with the primary purpose of characterizing the vertical variations of $\delta^{13}\text{C}$ across the Wasia Formation. All values are reported in per-mil relative to V-PDB (Vienna PeeDee Belemnite). Analytical reproducibility for both O and C isotopes is below 0.1‰.

Sedimentological evolution the Wasia Formation

Safaniya and Maaddud Members. The Safaniya Member consists of skeletal wackestones with benthic foraminifera (orbitolinids, trocholinids – [Figure 2](#)), suggesting a shallow marine setting. A marl layer with detrital material is present, followed by an increase in planktonic foraminifera and mudstones, indicating a deepening trend. The Middle Safaniya has high gamma-ray values, likely due to increasing organic content, and likely reflects anoxic conditions. The uppermost part of the Safaniya shows alternations of marl and

limestone beds suggesting increasing runoffs and detrital influxes. The Mauddud Member is divided into two units. The Lower Mauddud has aggrading skeletal wackestones with benthic foraminifera (mainly miliolids and *Praealveolina* spp.), dasycladacean debris, and gastropods, indicating a low-energy platform top setting. The Upper Mauddud displays an upward cleaning trend, with detrital-rich shales passing to skeletal wackestones and packstones, and rare rudist fragments, suggesting a broad deposition within a low to moderate-energy platform top setting. The top of the Mauddud Member is dominated by rudist floatstones, in line with regional observations (e.g., [van Buchem et al. 2002](#)).

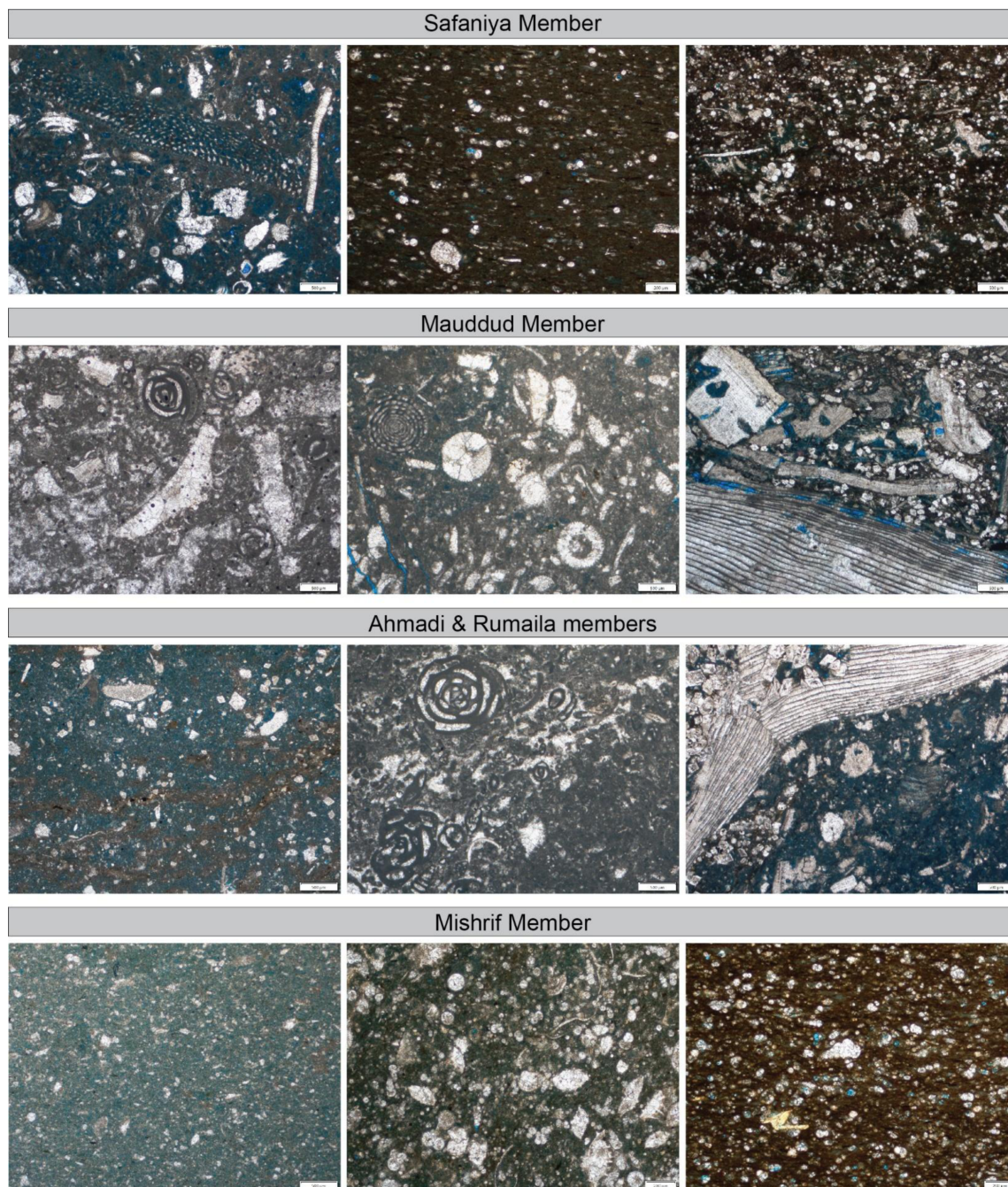


Figure 2—Thin-section photomicrographs illustrating the key microfacies observed in the various members of the Wasia Formation.

Wara, Ahmadi and Rumaila Members. The Wara Member is characterized by thick brown to greenish marls dominated by detrital clastic material, interbedded with skeletal wackestones and wacke-

packstones. Bioturbation is common, and the floro-faunal assemblage is dominated by gastropods, minor benthic foraminifera, green algae debris, and bivalve shells. The sedimentary patterns suggest a deposition within a low-energy, clay-prone restricted platform-top/inner ramp setting. The Ahmadi Member encompasses a broad upward coarsening trend, with pervasively bioturbated wackestones passing upwards to skeletal-rich wackestones and wacke-packstones with minor to common rudist debris. The benthic foraminifera assemblage is diverse and abundant, comprising large benthic foraminifera, including alveolinids, chrysalidinidae, and trocholinidae (Figure 2). The floro-faunal assemblage includes green algae debris, gastropods, solitary corals, echinoderms, and bivalve debris. The matrix of the floatstones is typically mud-dominated, suggesting that rudists developed in a low to moderate-energy platform-top setting. The Rumaila Member also accounts for an upward coarsening trend, similar to the Ahmadi Member. The basal section is characterized by pervasively bioturbated skeletal wackestones with local partial dolomitization (Figure 2). The skeletal assemblage is dominated by green algae debris, echinoderms, gastropods, bivalves, and benthic foraminifera. The uppermost part of the Rumaila Member is marked by a drop in the gamma-ray and is associated with the presence of rudist floatstones and rudstones, dominated by radiolitids. The sedimentological observations suggest a broad deposition within low-energy to moderate-energy platform top settings, possibly evolving upwards to rudist-shoal complexes at the top of the studied section.

Mishrif Member. The Mishrif Member is characterized by two distinct sections. The lowermost unit is marked by aggrading skeletal wackestones interbedded with laminated brown marl beds, that are partially dolomitized. The skeletal allochems include green algae debris, gastropods, benthic foraminifera (eg. miliolids, *Praealveolina* spp., *Nezzazata* spp), and rare planktonic foraminifera. This section likely represents the initial flooding on top of the Rumaila Member, with deposition in a low-energy, locally clay-rich inner ramp to mid-ramp environment. The upper section of the Mishrif Member consists of skeletal mudstones and wackestones with rare bivalve floatstones. Two "dirtying" upward trends are observed, with increasing gamma-ray values associated with increasing organic matter content. Planktonic foraminifera, pelagic crinoids, and phosphatic debris are present, indicating deposition in a deep marine setting, but slightly shallower than the source rock deposits of the Natih B in Oman (Al Balushi & Macquaker, 2011). The intervals with lower gamma-ray values are characterized by skeletal wackestones with echinoderm debris and minor benthic foraminifera, suggesting mid-ramp/slope deposits.

Carbon isotope data

The composite section shows positive $\delta^{13}\text{C}$ values with no negative excursions (Figure 3). A prominent positive excursion (+4‰) occurs in the lower Safaniya Member, followed by decreasing values to +1‰ in the middle Safaniya. The Ahmadi and Rumaila members have positive excursions, with the heaviest values (+4.8‰) in the Ahmadi Member. The Mishrif Member has varying $\delta^{13}\text{C}$ values, with a decrease to +2‰ in deep marine deposits and an increase to +4.2‰ in shallower conditions. The values then decrease back to +2‰, before increasing again to +3.5‰ at the top of the Mishrif Member.

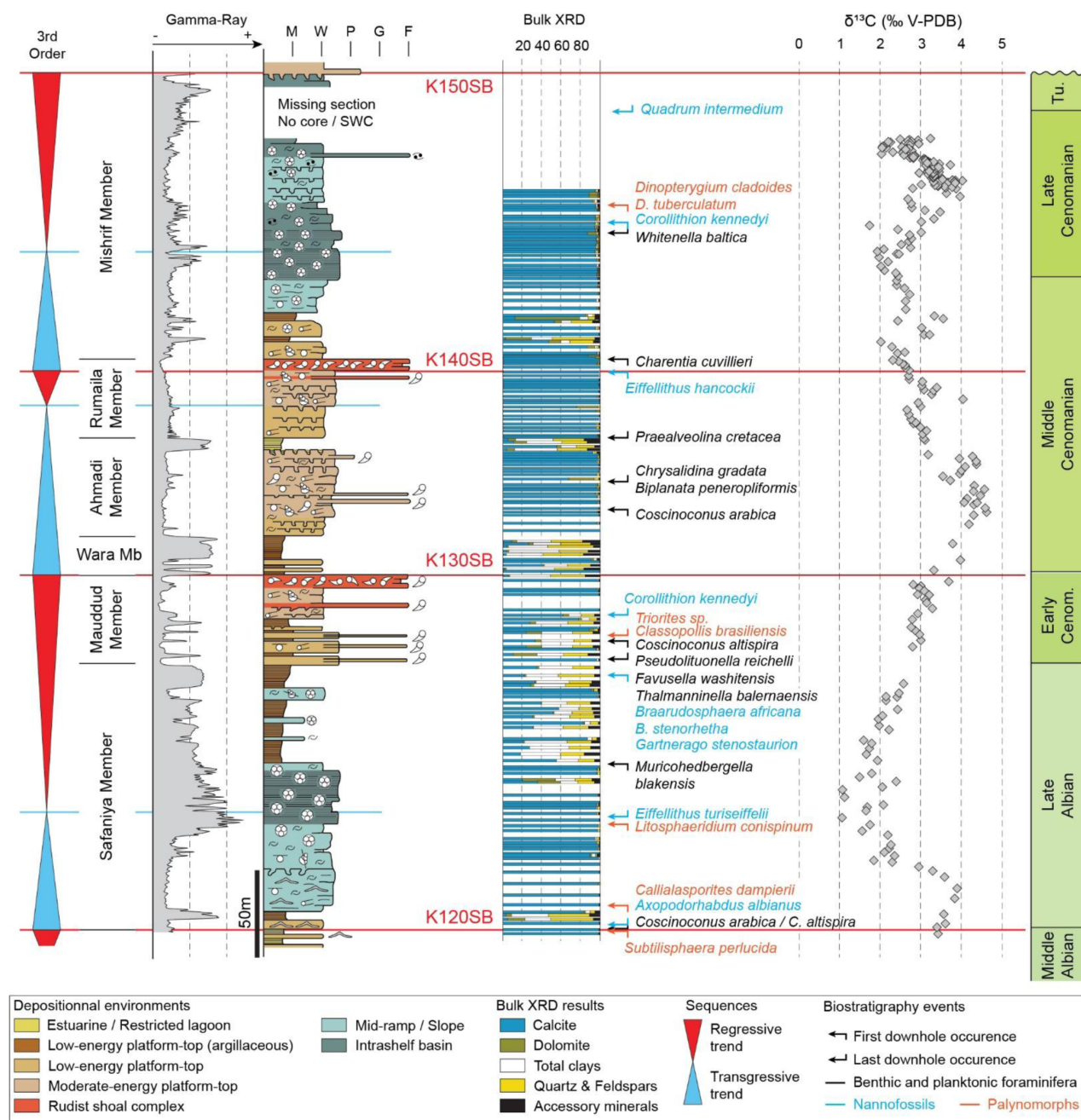


Figure 3—Composite section of the Wasia formation, including sequence stratigraphy, bulk XRD data, key bioevents, carbon isotope data and interpreted age dating of the section.

Discussion

The age dating of Albian/Turonian carbonate of the Arabian Platform is mainly based on ammonite biostratigraphy with limited biostratigraphy based on benthic foraminifera, rudists, and other macrofossil groups (e.g. Philip et al., 1995; van Buchem et al., 2002; Vicedo et al., 2013; Piuz et al., 2014; Meister & Piuz, 2015; Vahrenkamp et al., 2015; Vincent et al., 2015; Piuz & Vicedo, 2020). Palynology data are very rare and the limited nannofossil data often provide conflicting results with the other groups (see Homewood et al., 2008; Hennhoefler et al., 2018). Carbon isotopes have been used for platform-scale correlations, but show conflicting results due to limited biostratigraphic calibration (e.g. Vahrenkamp, 2013; Vahrenkamp et al., 2015; Wohlwend et al., 2016; Hennhoefler et al., 2018; Navidtalab et al., 2024). This study aims to

refine the biostratigraphy of the Albian/Turonian intervals using key bioevents and integrate it with carbon isotope data to establish a sequence stratigraphic framework (Figure 3).

Integrated biostratigraphy and carbon isotope stratigraphy

In the Safaniya Member, the last downhole occurrence (LDO) of the benthic foraminifera *Coscinoconus arabicus* and *C. altispira* indicates the base of the Late Albian, which broadly coincides with the top of the Khafji Member. This boundary also corresponds to the FDO of the dinocyst *Subtilisphaera perlucida*, which along with the LDO of the coccolith *Axopodorhabus albianus*, marks the top of the Middle Albian (Figure 3). The FDO of *Calliasporites dampierii* and *Litosphaeridium conispinum*, along with the LDO of the nannofossil *Eiffellithus turiseiffelii*, are Late Albian bioevents (Figure 3). Within the marl beds of the uppermost Safaniya Member, various amounts of *Calliasporites dampierii* and nannofossils such as *Braarudosphaera africana* and *B. stenorthetha*, are present, which are all restricted to the Albian. Disaggregated planktonic foraminifera in the Safaniya Member also indicate the FDO of *Thalmanninella balernaensis* in the uppermost limestone beds of the Safaniya (figure 3), which is restricted to the Albian in the Tethys (Bidgood & Simmons, 2022), suggesting that the Safaniya/Mauddud boundary broadly marks the top Albian. The base of the Mauddud Member hosts benthic foraminifera such as *Pseudolituonella reicheli*, *Praealveolina* spp., which indicate a Cenomanian or younger age, but are not found in the Albian. This is clear evidence that the Mauddud Member in this area is Cenomanian in age. In the Lower Mauddud, the presence of the pollen *Triorites* sp., and of possible *Classopolis brasiliensis*, however rare, are clear indicators of a Cenomanian age for the Lower Mauddud (Figure 3). The FDO of *Coscinoconus arabicus* in the Ahmadi Member along with the LDOs of large benthic foraminifera such as *Biplanata peneropliformis*, *Chrysalidina gradata*, and to some extent *Praealveolina cretacea*, pinpoint the Early to Mid-Cenomanian boundary. The LDO of the nannofossil *Corollithion kennedyi* in the Mauddud, associated with the FDO of *Eiffellithus hancockii* at the top Rumaila, suggests a broad Early to Middle Cenomanian age for this interval. The FDO of the benthic foraminifera *Charentia cuvillieri* at the top of the Rumaila is used as a marker for the top Middle Cenomanian. Within the deep marine facies of the uppermost Mishrif, rare occurrences of *Whitenella baltica* support a late Cenomanian age (Bidgood & Simmons, 2022; Navidtalab et al., 2024). In the Mishrif Member, the FDO of *Dinopterigium cladoides/tuberculatum* (dinocysts), along with the nannofossils *Corollithion kennedyi*, are intra Late Cenomanian events. Towards the top of the Mishrif Member, the LDO of *Quadrum intermedium*, restricted to the Turonian, marks the Cenomanian/Turonian transition. This suggests that the uppermost part of the Mishrif Member is of Late Cenomanian to Turonian and that most of the Turonian deposits (equivalent to Natih A in Oman) are likely missing.

In the Aruma section, the presence of the large benthic foraminifera *Pseudochubbina* spp., along with the continental palynomorph *Tubistephanocolporites cylindricus*, suggest a Campanian age for this interval, implying a ~10Ma time gap comprised in the Pre-Aruma Unconformity, which is associated with dissolution and cementation (Deville de Periere & Mangelot, 2025).

Carbon isotope data from the Arabian Platform are used to refine the stratigraphy of Albian to Turonian strata. The integrated biostratigraphy data from the Rub al Khali Basin are used to refine carbon isotope correlations both within the Arabian Platform and with key localities in the Tethys. The carbon isotope curve shows similar patterns and events that can be traced across the platform (e.g., Vahrenkamp, 2013; Vahrenkamp et al., 2015; Vincent et al., 2015; Wohlwend et al., 2016; Hennhofer et al., 2019; Navidtalab et al., 2024), including the positive excursion observed within the lower Safaniya Member, which corresponds to the OAE1d (Breistroffer Event).

The presence of Albian nannofossils, palynomorphs and well-preserved planktonic foraminifera in the Upper Safaniya suggests a thicker Late Albian sequence than expected. Two positive $\delta^{13}\text{C}$ excursions are recorded within the Ahmadi and Rumaila members, which coincide with those measured in the subsurface Natih D and Natih C, and correspond to the Middle Cenomanian Event 1 (e.g., Jarvis et al., 2006; Vahrenkamp, 2013; Wohlwend et al., 2016). The $\delta^{13}\text{C}$ values progressively increase upward during the late

Cenomanian until the onset of the Cenomanian/Turonian Boundary Event (OAE2) marked by a positive excursion.

Sequence stratigraphy

Three large-scale sequences (3rd order) were identified in the study area (Figure 3). The lowermost sequence (K120) corresponds to the Safaniya and Mauddud Members, with the MFS located in the Albian organic-rich deposits of the middle Safaniya Member. The middle sequence (K130) comprises the Wara, Ahmadi, and Rumaila members, with the K130 MFS interpreted within the low-energy inner ramp facies of the Rumaila Member. The uppermost sequence (K140) corresponds to the Mishrif Member, with the MFS located in the deepest and most distal section, broadly coinciding with the middle/late Cenomanian boundary. The K140 sequence is bounded by the K150 SB, which marks a significant event of erosion and exposure, dated to the middle Turonian (Simmons et al., 2024). The K150 SB marks the Pre-Aruma Unconformity, which has eroded a significant portion of the Turonian section, resulting in a ~10Ma time gap. The overlying Aruma section has a Campanian age, indicating a major hiatus in the sedimentary record. The K150SB is a prominent sequence boundary, characterized by karstification and cementation (Deville de Periere & Manguot, 2025).

Conclusions

This study integrates sedimentology, biostratigraphy, and carbon isotope data to improve the stratigraphic understanding of the Wasia Formation in the Rub al Khali Basin, Saudi Arabia. The results provide a better time constraint on the facies changes and sequence stratigraphy, which reflect platform-scale sea level fluctuations. Three sequences were identified: K120 (Safaniya and Mauddud members), K130 (Wara, Ahmadi, and Rumaila members), and K140 (Mishrif Member). Each sequence shows distinct facies changes and biota modifications, with key events including the Late Albian OAE1d, the Mid-Cenomanian Event 1 and the Cenomanian/Turonian boundary event (OAE2). The study suggests that the regional age dating of the Safaniya Member may need revision, and that most of the Turonian deposits are missing due to erosion by the Pre-Aruma Unconformity, resulting in a ~10Ma stratigraphic gap. The integrated approach provides a more accurate understanding of the stratigraphy and age dating of the Wasia Formation, and highlights the importance of considering multiple fossil groups and carbon isotope data in sequence stratigraphic analysis.

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